PYTHON CODING EXERCISE

Scenario: Build a small Python application that processes job execution records.

Input: Each record contains: job_id, status, started_at, finished_at, and attempt.

Requirements

- Read records from a JSON file.
- Validate the input and handle malformed records gracefully.
- Calculate execution duration for completed jobs.
- Produce a summary containing: total jobs, successful jobs, failed jobs, average successful-job duration, and jobs that required retries.
- Structure the solution as you would production code rather than as a single monolithic script.
- Include automated tests for important behavior.
- Include a short README explaining how to run the program/tests and documenting key assumptions or design decisions.

Tools: Standard Python libraries and reasonable third-party packages are allowed. Normal development tools, including AI coding assistants, may be used.

Expected effort: No more than 60–90 minutes. We are not looking for a polished production system.